

Tower of Hanoi Software Development Schedule (10 Weeks)

Project Information

Project Name: Tower of Hanoi Software

Programming Language: Java

Course: Software Development 1

Week 1: Project Planning and Requirement Analysis

What I Will Do

- Analyze the project requirements.
- Study the Tower of Hanoi problem and its rules.
- Identify software features and modules.
- Prepare project scope and objectives.
- Create software development roadmap.

What I Need to Learn

- Introduction to Software Development Life Cycle (SDLC)
- Tower of Hanoi Algorithm
- Project Requirement Analysis
- Git and GitHub Basics

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- Project Proposal
- Requirement Analysis Document
- Initial README File

What I Will Demonstrate

- Project Idea
- Software Features
- Project Objectives
- Development Plan

Week 2: Software Design

What I Will Do

- Design software architecture.
- Create class diagram.
- Design software workflow.
- Plan user interface layout.

What I Need to Learn

- Object-Oriented Programming (OOP)
- UML Class Diagram
- Java Classes and Objects
- Software Design Principles

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- System Design Document
- Class Diagram
- Project Structure

What I Will Demonstrate

- Software Architecture
- Class Diagram
- Workflow Design

Week 3: User Interface Development

What I Will Do

- Create application window.
- Develop Main Menu.
- Add Start, Help and Exit options.
- Design navigation system.

What I Need to Learn

- Java Swing
- JFrame
- JButton
- Event Handling

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- Main Menu Source Code

- GUI Design Files

What I Will Demonstrate

- Working Main Menu
- User Navigation System

Week 4: Game Board Development

What I Will Do

- Create three towers.
- Create disks with different sizes.
- Display game board visually.
- Improve interface appearance.

What I Need to Learn

- Java Graphics
- JPanel
- Drawing Components
- GUI Layout Management

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- Tower Module
- Disk Visualization Module

What I Will Demonstrate

- Software Interface
- Tower and Disk Visualization

Week 5: Gameplay Implementation

What I Will Do

- Implement disk movement.
- Allow user interaction.
- Update game board after every move.

What I Need to Learn

- Mouse Events
- Java Collections (Stack)
- Event Driven Programming

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- Gameplay Logic
- Disk Movement Module

What I Will Demonstrate

- User Moving Disks
- Basic Working Software

Week 6: Rule Validation System

What I Will Do

- Prevent illegal moves.
- Validate every move.
- Display warning messages.

What I Need to Learn

- Conditional Logic
- Error Handling
- Validation Techniques

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- Validation Module
- Error Handling Module

What I Will Demonstrate

- Rule Checking System
- Invalid Move Detection

Week 7: Software Features Enhancement

What I Will Do

- Add Move Counter.
- Add Game Timer.
- Add Restart Option.
- Improve software usability.

What I Need to Learn

- Java Timer
- Labels and Dynamic Updates
- State Management

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- Timer Feature
- Move Counter Feature
- Restart Function

What I Will Demonstrate

- Timer Working
- Move Counter
- Restart Functionality

Week 8: Auto Solver Development

What I Will Do

- Implement recursive Tower of Hanoi solution.
- Create Auto Solve button.
- Show solution step-by-step.

What I Need to Learn

- Recursion
- Recursive Algorithms
- Algorithm Visualization

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- Recursive Algorithm Module
- Auto Solver Feature

What I Will Demonstrate

- Automatic Puzzle Solving
- Recursion Implementation

Week 9: Software Finalization

What I Will Do

- Add Win Detection.
- Add Success Message.
- Improve UI Design.
- Add About Section.

What I Need to Learn

- Software Testing Basics
- UI Improvement Techniques
- User Experience Design

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- Win Detection Module
- Updated User Interface
- About Section

What I Will Demonstrate

- Complete Software Workflow
- Winning Screen
- Improved Interface

Week 10: Testing and Final Release

What I Will Do

- Perform complete testing.

- Fix bugs and errors.
- Prepare documentation.
- Prepare final presentation.

What I Need to Learn

- Software Testing
- Debugging
- Documentation Writing
- Presentation Skills

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- Final Source Code
- Documentation
- User Manual
- Screenshots
- Version 1.0 Release

What I Will Demonstrate

- Complete Software Application
- All Features Working
- Final Project Presentation